

Beverage Sales Forecasting

1. Problem Statement

Build a forecasting system to predict the next 8 weeks of beverage sales for 43 US states using historical weekly sales data from 2019 to 2023.

2. Dataset Overview

Field	Details
File	Forecasting_Case-Study.xlsx
Total Rows	8,084
Columns	State, Date, Total, Category
States	43 US States
Date Range	January 2019 - December 2023
Category	Beverages
Missing Values	None
Frequency	Weekly (irregular)

3. Exploratory Data Analysis (EDA)

- Loaded and explored dataset structure — 8,084 rows, 4 columns
- Converted Date column to proper datetime format
- Visualized weekly sales trend for California — identified clear upward trend and seasonality
- Top 5 states by total sales: Texas, California, Florida, New York, Georgia
- Data shows COVID-19 impact in 2020-2021 with recovery spikes

4. Feature Engineering

12 features were created from raw data to enhance model performance:

Feature	Description
lag_1	Sales from 1 week ago
lag_4	Sales from 4 weeks ago
lag_8	Sales from 8 weeks ago
rolling_mean_4	4-week rolling average
rolling_std_4	4-week rolling standard deviation

rolling_mean_8	8-week rolling average
week	Week number of year
month	Month number (1-12)
quarter	Quarter of year (1-4)
year	Year
dayofweek	Day of week (0-6)
is_holiday	US Holiday flag (0 or 1)

5. Train / Test Split

Split	Date Range	Rows
Train	2019-11-17 to 2023-04-23	7,396
Test	2023-04-30 to 2023-12-03 (last 8 dates)	344

6. Models Trained

6.1 ARIMA

- Trained separately for each of 43 states
- Parameters: order=(1,1,1)
- MAE: 11,477,888 | RMSE: 13,158,763

6.2 Prophet (Facebook)

- Trained separately for each of 43 states
- Yearly + Weekly seasonality enabled
- MAE: 13,983,471 | RMSE: 17,494,146

6.3 XGBoost — BEST MODEL

- Trained separately for each of 43 states
- Parameters: n_estimators=100, learning_rate=0.1, max_depth=4
- Used all 12 engineered features
- **MAE: 10,935,476 | RMSE: 13,225,062**

6.4 LSTM (PyTorch)

- Sequence length: 8 weeks
- Architecture: 2 LSTM layers, hidden size 64
- Trained 50 epochs per state
- MAE: 11,750,329 | RMSE: 13,447,228

7. Model Comparison

Rank	Model	MAE	RMSE
1 ✓BEST	XGBoost	10,935,476	13,225,062
2	ARIMA	11,477,888	13,158,763
3	LSTM	11,750,329	13,447,228
4	Prophet	13,983,471	17,494,146

Winner: XGBoost — Lowest MAE across all 43 states. 43 individual state models saved using Joblib.

8. REST API — FastAPI

Endpoint	Method	Description
/	GET	Health check — API status
/states	GET	Returns list of all 43 available states
/forecast	POST	Returns 8-week sales forecast for given state

Sample API Request

POST /forecast

```
{
  "state": "California",
  "start_date": "2024-01-01",
  "lag_1": 850000000,
  "lag_4": 820000000,
  "lag_8": 800000000,
  "rolling_mean_4": 830000000,
  "rolling_std_4": 15000000,
  "rolling_mean_8": 825000000
}
```

Sample API Response

```
{
  "state": "California",
  "model": "XGBoost",
  "forecast_weeks": 8,
  "predictions": [
    {
      "date": "2024-01-01",
      "predicted_sales": 824752896
    },
    {
      "date": "2024-01-08",
      "predicted_sales": 840124160
    },
    {
      "date": "2024-01-15",
      "predicted_sales": 824175616
    },
    ... (8 weeks total)
  ]
}
```

9. Tech Stack

Tool	Purpose
Python 3.12	Core programming language
Pandas, NumPy	Data loading and processing
Scikit-learn	Preprocessing and evaluation metrics
Statsmodels	ARIMA model
Prophet	Facebook Prophet forecasting model
XGBoost	Best performing gradient boosting model
PyTorch	LSTM deep learning model
FastAPI + Uvicorn	REST API server
Holidays	US holiday feature generation
Joblib	Model serialization and saving
Matplotlib, Seaborn	Data visualization
Jupyter Notebook	Development environment

10. Project Structure

```

Forecasting_Project/
├── data/
│   └── Forecasting_Case-Study.xlsx
├── notebooks/
│   └── forecasting.ipynb ← EDA + All 4 Models + Evaluation
├── src/
│   └── api.py ← FastAPI REST API
├── models/
│   └── xgb_*.pkl ← 43 saved XGBoost models
├── requirements.txt
└── README.md

```

11. Setup Instructions

Step 1 — Clone repository:

```

git clone https://github.com/ankita-2021/Beverage-Sales-Forecasting.git
cd Forecasting_Project

```

Step 2 — Create virtual environment:

```

py -3.12 -m venv venv
venv\Scripts\activate

```

Step 3 — Install dependencies:

```

pip install -r requirements.txt

```

Step 4 — Run API:

```
uvicorn src.api:app --reload
```

Step 5 — Open Swagger UI in browser:

`http://127.0.0.1:8000/docs`

12. GitHub Repository

<https://github.com/ankita-2021/Beverage-Sales-Forecasting>